

Toluwani Samuel Aremu

AI Safety & Security

[Certifications](#) • [Email](#) • [GitHub](#) • [Google Scholar](#) • [LinkedIn](#) • [Website](#)

PROFESSIONAL SUMMARY

AI Safety and Security researcher specializing in trustworthy generative AI, watermarking, adversarial robustness, and monitoring systems. PhD researcher at MBZUAI with publications across ICML, ACL, CVPR, IJCAI, SaTML, and AAAI, as well as spotlight/distinguished-paper awards. Experienced in adapting across domains, from privacy-preserving ML and AI governance to agentic systems, generative AI platforms, and applied AI product development.

TECHNICAL SKILLS

- **AI Safety & Security:** [Threat Modeling](#) | [Watermarking](#) | [Scalable Oversight](#) | [Monitoring](#) | [Adversarial/Jailbreak Robustness](#) | [Governance](#)
- **Applied Systems:** [Agentic Systems](#) | [Generative AI Platforms](#) | [AI Product Prototyping](#) | [AI Applications in Financial Markets](#)
- **Tools:** [Python](#) | [PyTorch](#) | [Guardrails](#) | [vLLM](#) | [AutoGen](#) | [Unsloth](#) | [Antigravity](#) | [Codex](#) | [Claude Code](#)

EDUCATION

- Machine Learning (PhD) | **Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), UAE** AUG 2023 – PRESENT
 - **Advisor:** [Nils Lukas](#)
 - **Thesis:** [Robust Watermarking and Monitoring for Trustworthy Generative AI Systems](#)
 - **Expected Completion Date:** [May 2027](#)
- Machine Learning (MSc) | **Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), UAE** JAN 2021 – JAN 2023
 - **Advisor:** [Karthik Nandakumar](#)
 - **Thesis:** [Activation-Free Neural Networks for Efficient Private Inference](#)
- Computer Science (MSc) | **University of Ibadan (UI), Nigeria** MAY 2018 – MAY 2020
- Computer Science (BSc) | **Adeleke University, Nigeria** OCT 2012 – JUL 2016
- Theory and Practice of Music (Diploma) | **Associated Board of the Royal Schools of Music (ABRSM), UK** AUG 2012 – AUG 2015

EXPERIENCE

- Visiting Research Scientist | **Center for Frontier AI Research (CFAR), A*STAR** APR 2026 – JUL 2026
 - **Research Areas:** [Watermarking](#), [Threat Modeling](#)
 - **Host:** [Jie Zhang](#)
- AI Scientist (Quant/Treasury) | [Kora](#) FEB 2026 – PRESENT
 - I am spearheading the technical vision and engineering roadmap for our startup (incubated under Kora) dedicated to building and maintaining an internal hedge using self-improving agentic systems that digest complex market feeds and unstructured data, detect market regimes, code and backtest strategies, manage portfolios, and maximize alpha.
- AI Platform Developer (Contract) | [Vadis Media](#) MAY 2025 – JUL 2025
 - Engineered a generative AI [platform](#) connecting movie production companies with the right financiers that integrates multimodal agentic workflows for idea generation, actor recommendation, FX scenes rendering, product/brand placements, and AI-assisted cost and ROI estimation, successfully translating complex entertainment-finance workflows into automated features that reduce manual proposal preparation effort by 95% and shorten planning timelines from years to days.
- Doctoral Researcher | [Spot-AI Lab](#), MBZUAI AUG 2023 – PRESENT
 - **Research Areas:** [Watermarking](#), [Monitoring](#), [Threat Modeling](#)
- Research Engineer | [MCR Lab](#), MBZUAI FEB 2023 – AUG 2023
 - **Research Areas:** [Generative AI Applications](#)
- Applied Scientist (Intern) | **M42, UAE** FEB 2023 – APR 2023
 - **Project:** Health-Tech Reproducibility and Reporting Toolkit
- Graduate Research Assistant | [SPrint-AI Lab](#), MBZUAI JAN 2021 – JAN 2023
 - **Research Areas:** [Privacy-Preserving ML](#)

SELECTED PUBLICATIONS (* denotes equal contribution, ** denotes working paper, 🏆 denotes Spotlight/Award)

- **[**]** Aremu, Toluwani, Nils Lukas, and Jie Zhang. "Watermarking Should Be Treated as a Monitoring Primitive." arXiv preprint arXiv:2605.13095 (2026).
- **[**]** Aremu, Toluwani, Daniil Ognev, Samuele Poppi, and Nils Lukas. "Robust Safety Monitoring of Language Models via Activation Watermarking." arXiv preprint arXiv:2603.23171 (2026).
- **[CVPR-AdvML 🏆]** Fares, Samar*, Klea Ziu*, **Toluwani Aremu***, Nikita Durasov, Martin Takáč, Pascal Fua, Karthik Nandakumar, and Ivan Laptev. "Mirrorcheck: Efficient adversarial defense for vision-language models." The 6th Workshop of Adversarial Machine Learning on Computer Vision: Safety of Vision-Language Agents. 2026.
- **[**]** Aremu, Toluwani, Noor Hussein, Munachiso Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Gong, and Nils Lukas. "Mitigating Watermark Forgery in Generative Models via Randomized Key Selection." arXiv preprint arXiv:2507.07871 (2025).
- **[ICML 🏆]** Diaa, Abdulrahman, **Toluwani Aremu**, and Nils Lukas. "Optimizing Adaptive Attacks against Watermarks for Language Models." In International Conference on Machine Learning, pp. 13546-13569. PMLR, 2025.
- **[AAAI]** Aremu, Toluwani, Oluwakemi Akinwehinmi, Chukwuemeka Nwagu, Syed Ishtiaque Ahmed, Rita Orji, Pedro Arnau Del Amo, and Abdulmotaleb El Saddik. "On the reliability of Large Language Models to misinformed and demographically informed prompts." Association for the Advancement of Artificial Intelligence (AAAI) AI Magazine 46, no. 1 (2025): e12208.
- **[CVPR 🏆]** Vayani, Ashmal, Dinura Dissanayake, Hasindri Watawana, Noor Ahsan, Nevasini Sasikumar,..., **Toluwani Aremu** et al. "All languages matter: Evaluating Imms on culturally diverse 100 languages." In Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 19565-19575. 2025.

- [\[IJCAI\]](#) Tastan, Nurbek, Samar Fares, **Toluwani Aremu**, Samuel Horvath, and Karthik Nandakumar. "Redefining contributions: shapley-driven federated learning." In Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, pp. 5009-5017. 2024.
- [\[SaTML\]](#) **Aremu, Toluwani**, and Karthik Nandakumar. "Polykervnets: Activation-free neural networks for efficient private inference." In 2023 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), pp. 593-604. IEEE, 2023.

OTHER EXPERIENCE

- **Projects**
 - Accented Speech Recognition | **Course Project (MBZUAI)** AUG 2021 – DEC 2021
 - Racial Bias Mitigation in Self-Supervised Face Recognition Architecture | **Course Project (MBZUAI)** JAN 2021 – MAY 2021
 - Gender Bias Mitigation in Word Embeddings | **Course Project (MBZUAI)** JAN 2021 – MAY 2021
 - **Teaching**
 - Teaching Assistant, Introduction to AI (AI1010) | **MBZUAI** AUG 2025 – DEC 2025
 - Teaching Assistant, Mathematical Foundations of AI (MTH701) | **MBZUAI** AUG 2024 – DEC 2024
 - Teaching Assistant, Object Oriented Programming (OOP) | **University of Ibadan** JAN 2019 – MAY 2019
 - **Talks/Presentations**
 - Mitigating Watermark Forgery Attacks in Generative Models via Multi-Key Watermarking | **MBZUAI Research Showcase, UAE** OCT 2025
 - Literature Review and Research Methodologies | **LyngualLabs, Nigeria** MAY 2025
 - Optimizing Adaptive Attacks against Content Watermarks for Language Models | **ICLR-WMARK Workshop, Singapore** APR 2025
 - PolyKervNets: Activation-Free Neural Networks for Efficient Private Inference | **SaTML, USA** FEB 2023
 - Ethical Perspectives of AI | **Department of Material Sciences, University of Denver, USA** JUL 2022
 - **Service (Reviewer | Area Chair | Program Chair | Organizer | Editor)**
 - **Conferences:** [AAAI](#) ('25, '26, '27), [CVPR](#) ('26), [NeurIPS](#) ('25, '26), [ICLR](#) ('25, '26), [ICML](#) ('25, '26), [AISTATS](#) ('25), [DLI](#) ('25)
 - **Workshops:** [AI4GOOD@ICML](#) ('26), [HRAIM](#) ('25), [SafeGenAI@NeurIPS](#) ('25), [WMark@ICLR](#) ('25)
 - **Journals:** [TMLR](#) ('26), [IEEE TPAMI](#) ('25), [IEEE Access](#) ('25), [AI Magazine](#) ('25), [QSI](#) ('25), [CHBAH](#) ('25)
 - **Leadership/Volunteering**
 - Resident Life Assistant | **MBZUAI Undergraduate Student Life** AUG 2025 – MAY 2027
 - Volunteering Member | **EVOLVE-MBZUAI** AUG 2024 – MAY 2027
 - Volunteering Member | **Spruce Up Your Space** AUG 2024 – MAY 2027
 - Associate Editor | **ML Department Research Blog** JAN 2024 – MAY 2027
 - Student Editor | **MBZUAI Research Blog** JAN 2024 – MAY 2027
 - Mentor | **Research Mentorship for African Research Enthusiasts** JAN 2021 – PRESENT
-

HONORS & AWARDS

- CVPR-AdvML'26 **Distinguished** Paper Award MAY 2026
- ICML'26 **Gold** Reviewer Award (Top 25%) MAY 2026
- MBZUAI Award of Recognition for Service and Contribution to the MBZUAI Alumni Body FEB 2026
- MBZUAI ICML Conference Travel Grant | **Value of Award: \$4,500** JUL 2025
- ICML'25 **Spotlight** Paper Award (Top 25%) JUL 2025
- CVPR'25 **Spotlight** Paper Award (Top 25%) APR 2025
- MBZUAI ICLR Conference Travel Grant | **Value of Award: \$4,500** APR 2025
- MBZUAI PhD Fully Funded Fellowship | **Value of Award: ~\$400,000** AUG 2023
- MBZUAI MSc Fully Funded Fellowship | **Value of Award: ~\$175,000** JAN 2021
- UAE **Golden** Visa for Talented Persons/Specialists in Science OCT 2022
- MBZUAI Award of Appreciation for Iconic Representation and Student Hospitality AUG 2022
- ProjectSet Innovation Challenge for Entrepreneurship (ICE-22) | **Value of Award: ~\$3,000 in Cloud Credits** MAY 2022
- Top 100, DeepLearning.AI Data-Centric AI Competition AUG 2021
- NYSC-FRSC Award for the Most Creative Corp Member OCT 2017
- 2015 AUE-NACOSS Award for the Best Student Programmer JUL 2015